

Civics Group	A Level Index Number	Name (use BLOCK LETTERS)
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H2



**ST. ANDREW'S JUNIOR COLLEGE
2022 JC2 PRELIMINARY EXAMINATIONS**

H2 BIOLOGY

9744/03

Paper 3 (Booklet A)

Thursday

15th September 2022

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

All working for numerical answers must be shown.

Conceptual error (C)	Data Quoting (D)	Expression (E)	Misreading the question (Q)

For Examiners' Use	
1	/30
2	/9
3	/11
4 or 5	/25
Total	/75

This document consists of **8** printed pages.

[Turn over]

Answer **all** questions.

QUESTION 1

- (a) SWI/SNF (SWItch/Sucrose Non-Fermentable) is a subfamily of ATP-dependent chromatin remodelling complexes found in eukaryotes. Products of SWI and SNF genes can destabilise histone-DNA interactions, resulting in sliding or ejection of nucleosomes, as shown in Fig. 1.1.

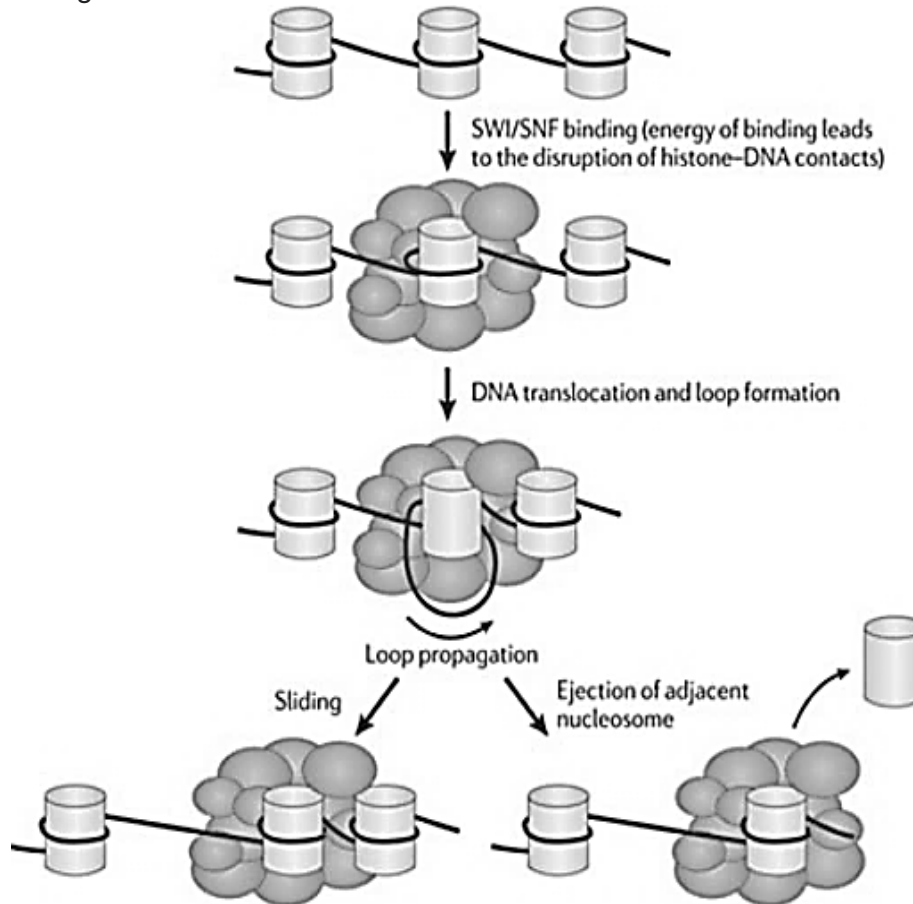


Fig. 1.1

In contrast, Polycomb Repressive Complex (PRC), with the histone methyltransferase EZH2 as its catalytic subunit, is associated with chromatin compaction.

- (i) With reference to Fig. 1.1, describe how the activity of SWI/SNF complexes can lead to the active transcription of genes.

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- (ii) Describe two mechanisms that can produce the same chromatin remodelling effect as PRC/EZH2 complex.

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- (iii) Describe how the nucleosomes will eventually pack into chromosomes during prophase of mitosis.

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- (b) Over 20% of human cancers carry a mutation in SWI/SNF complex subunit genes. Fig. 1.2 shows that the inactivation of SWI/SNF causes a failure to inhibit PRC at promoters and typical enhancers (TE) of genes involved in differentiation. The residual functional SWI/SNF complexes are preferentially localized to super-enhancers (SE) of genes maintaining cell renewal. The resulting imbalance between differentiation vs. self-renewal promotes tumorigenesis (tumour formation) .

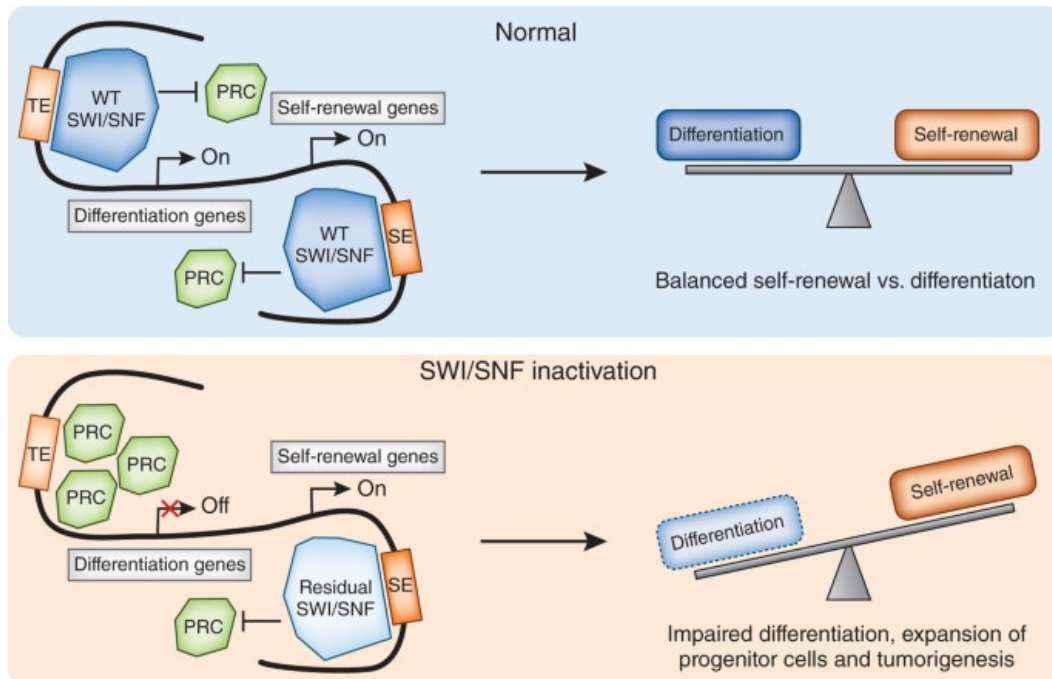


Fig. 1.2

- (i) Describe the role of enhancers in the control of gene expression.

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(ii) With reference to information from **1(a)** as well as Fig. 1.2, describe the downstream effect of failing to inhibit PRC due to inactivation of SWI/SNF.

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(iii) Explain how the “imbalance between differentiation vs. self-renewal promotes tumorigenesis”.

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- (c) Different mutations in the proto-oncogene coding for Epidermal Growth Factor Receptor (EGFR) are associated with several cancers such as lung cancer. Lung tissue samples were taken from two lung cancer patients and cultured on Petri dishes for 24 hours. The average cytoplasmic mRNA levels (Fig. 1.5A) as well as 6-hourly protein levels (Fig. 1.5B) of EGFR and β -actin were investigated. β -actin is a house-keeping gene that is expressed in almost all cell types and is used as a positive control.

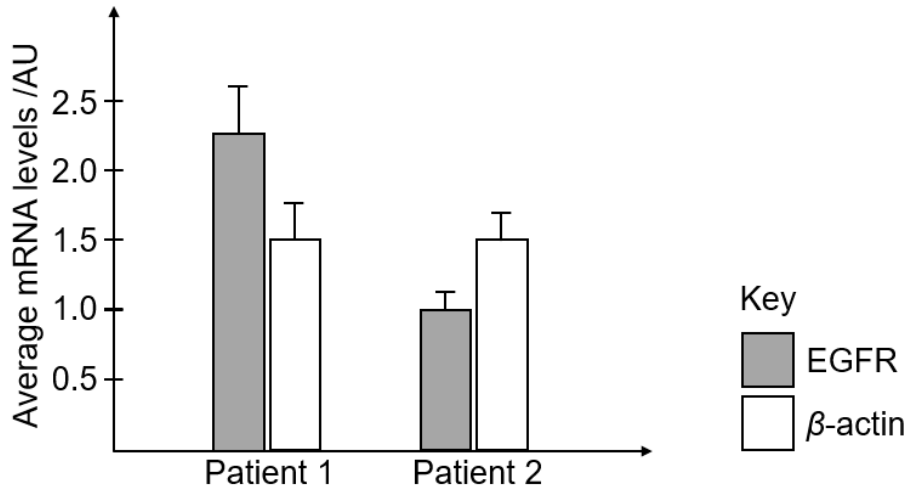


Fig. 1.3A

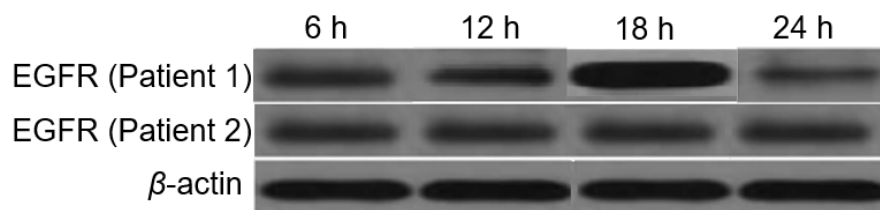


Fig. 1.3B

- (i) Describe the rationale of using β -actin as a positive control.

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- (ii) Describe the difference in average mRNA and protein level of EGFR between the two patients.

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- (iii) Suggest and explain possible mutations in the EGFR gene of patients 1 and 2 that could result in the differences described in **c(ii)**.

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- (iv) Contrast between the function of a proto-oncogene such as EGFR and a tumour suppressor gene such as p53.

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- (v) The tissue samples taken from the two lung cancer patients showed mutations in several other genes besides proto-oncogenes and tumour suppressor genes.

Explain how these mutations contributed to the development of lung cancer.

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[Total: 30]

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H2

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2022 JC2 PRELIMINARY EXAMINATIONS**

H2 BIOLOGY**9744/03****Paper 3 (Booklet B)**

Thursday

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For Examiners' Use	
2	/9
3	/11
4 or 5	/25
Total	/75

This document consists of **16** printed pages.

[Turn over]

QUESTION 2

Measles is an infectious disease for which vaccines have been developed. The commonly used vaccine consists of an attenuated (weakened) form of the virus. The measles vaccine is normally given to children when they are about one year old, followed by a booster dose when they are about four years old.

Fig. 2.1 shows the number of reported cases of measles and the percentage of the population vaccinated worldwide between 1980 and 2002.

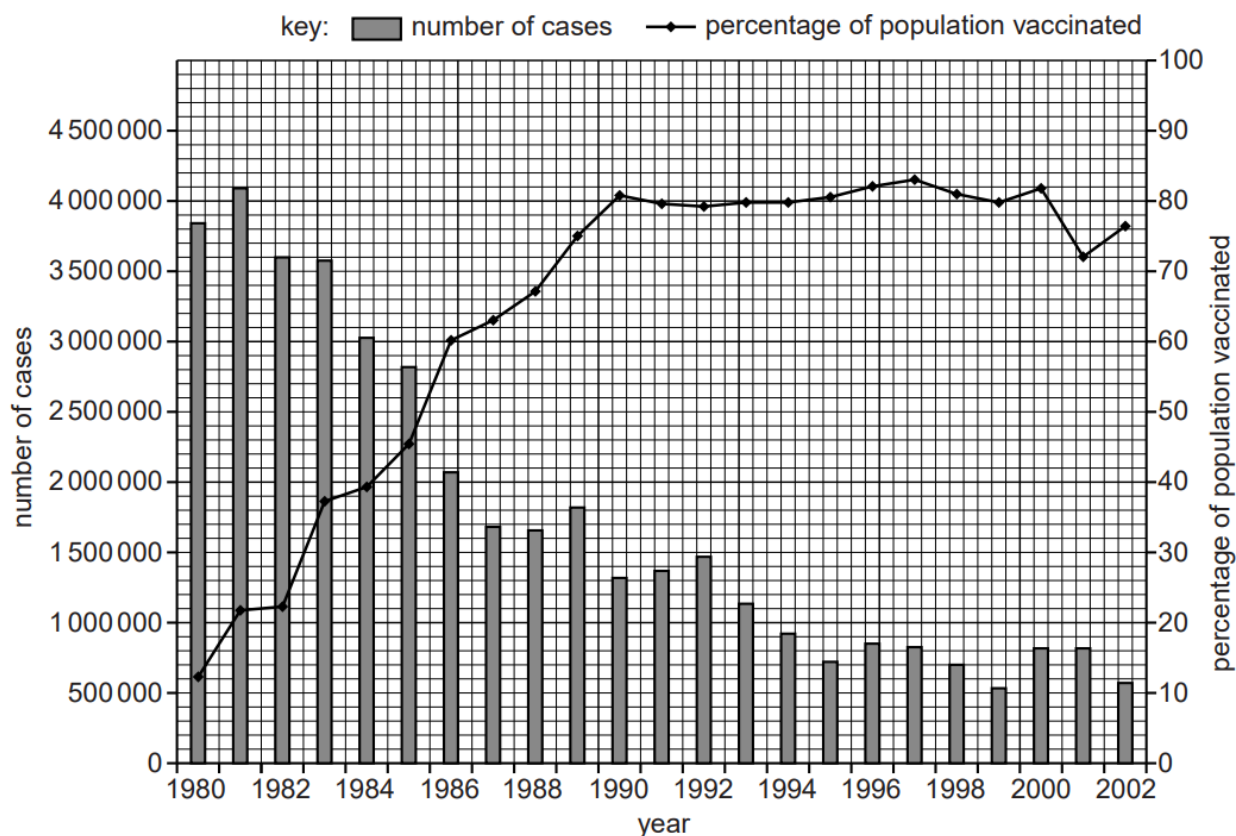


Fig. 2.1

(a) Describe the trends shown in Fig. 2.1:
between 1980 and 1990

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between 1990 and 2002.

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(b) The vaccination of a high enough percentage of the population can break the disease transmission cycle, giving 'herd immunity'.

The measles virus spreads easily by droplet transmission. One estimate of its reproduction number, R_0 , is 15, meaning that each person infected with measles is expected to infect 15 other people.

If the percentage of people who are immune to a disease exceeds the herd immunity threshold, the disease can no longer persist in the population. Assuming 100% efficacy of the vaccine, the herd immunity threshold is calculated as:

$$100 \times (1 - 1/R_0)$$

(i) Calculate the herd immunity threshold for measles, if $R_0 = 15$.

Herd immunity threshold =% [1]

(ii) Use Fig. 2.1 and your answer to **(b)(i)** to explain why the number of measles cases remain above 500,000.

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- (c) The measles virus has a unique protein on its surface called MV-H which can bind to a protein called CD-46 on the surface of human cells. This allows the measles virus to infect these cells.

Suggest how the two proteins, MV-H and CD-46, can bind to each other.

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[Total: 9]

QUESTION 3

Fig. 3.1 shows the percentage cover of live corals and the density of herbivorous fish (plant-feeding) on a coral reef over a number of years. Due to unusually warm water, many of the corals living on the reef died in 1998.

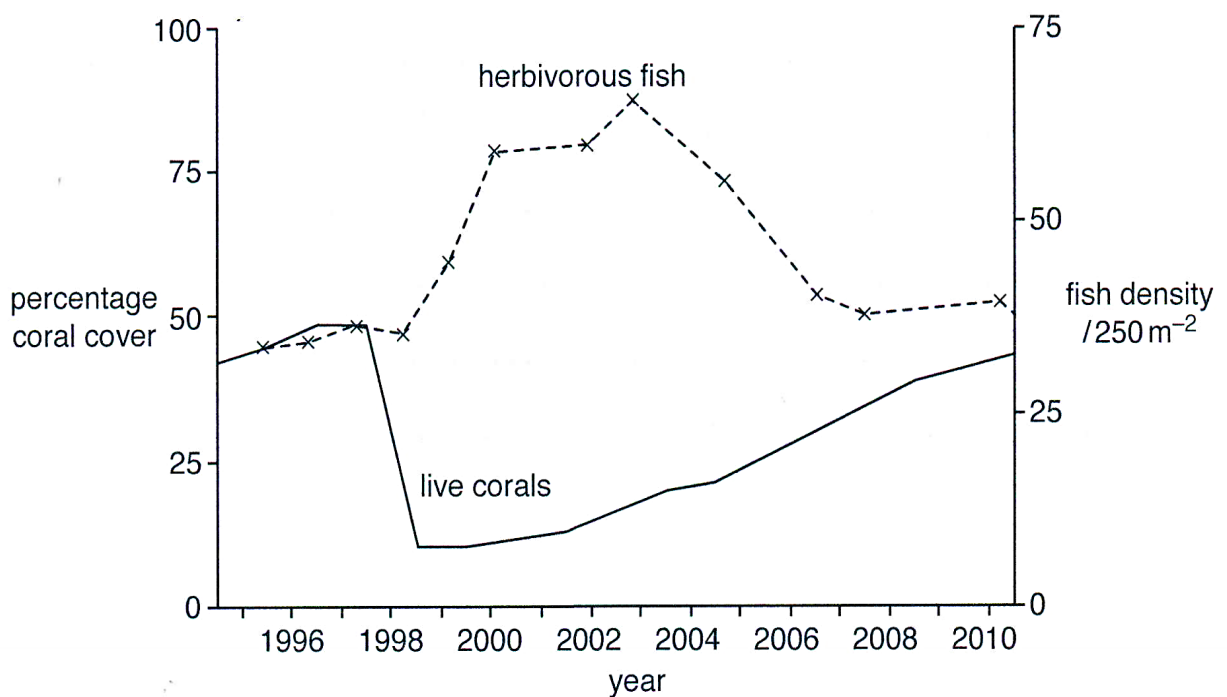


Fig. 3.1

(a) Describe the change in percentage cover of live corals shown in Fig. 3.1.

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(b) Suggest possible impacts due to the change in percentage cover of the live corals.

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(c) State a possible reason why the density of herbivorous fish increases after the death of many of the corals in 1998.

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The impact of climate change is a major threat not only for corals but also for life on earth a whole. Fig. 3.2a shows the loss of ice mass between 1992 and 2018. Fig. 3.2b shows the median extent of sea ice concentration in 1992 as compared to 2018. Fig. 3.3 the number of polar bears living in this biome.

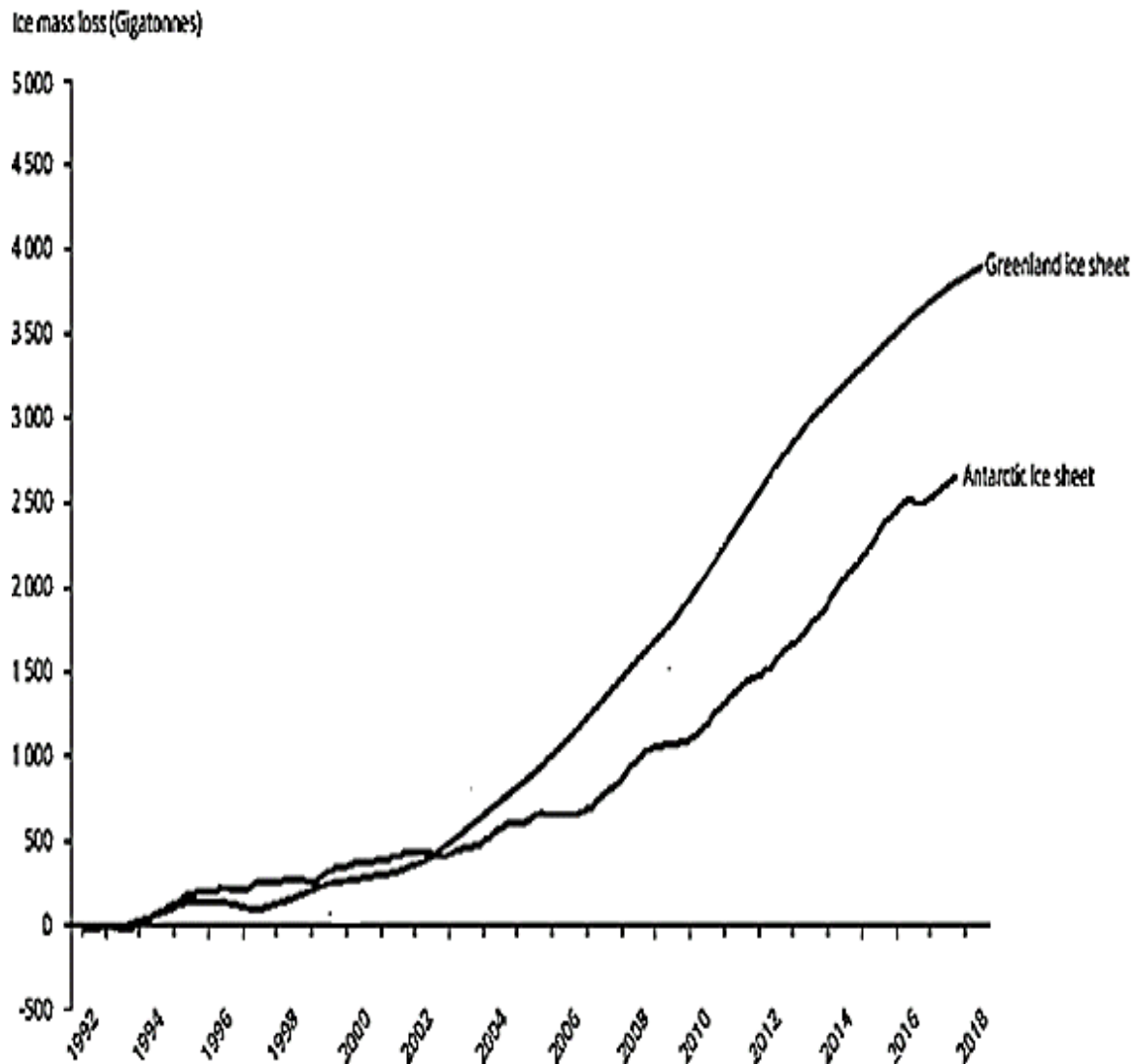


Fig. 3.2a

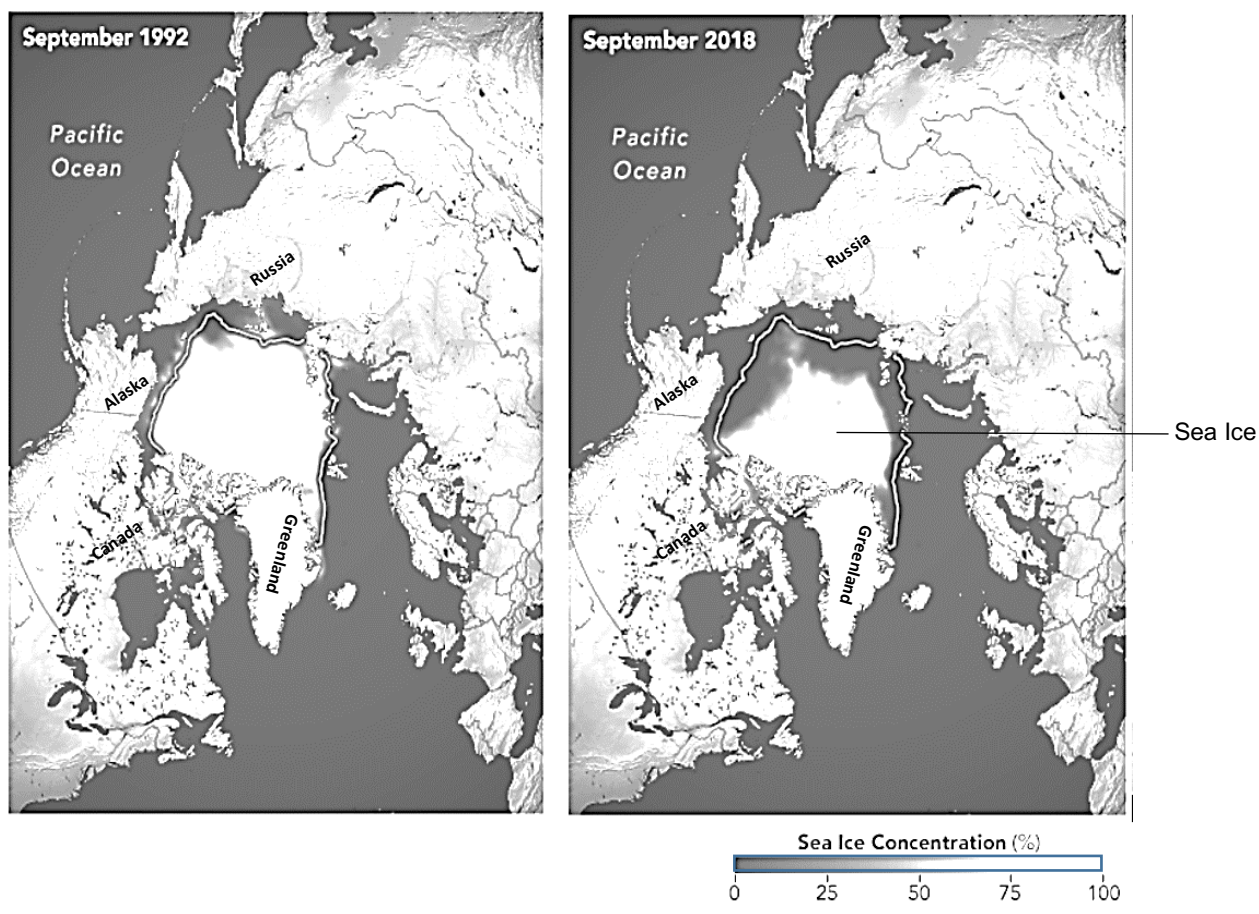


Fig. 3.2b

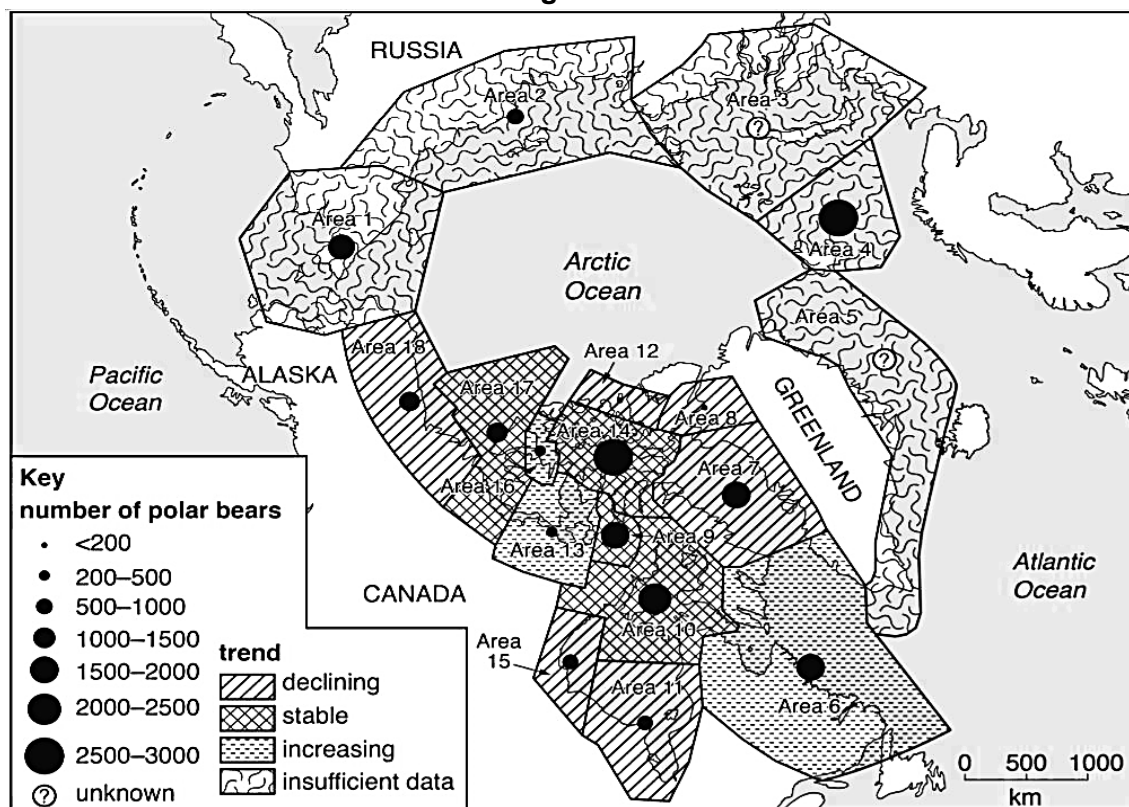


Fig. 3.3

(d) With reference to Figs. 3.2a, 3.2b and 3.3, describe and explain the evidence which suggest that polar bears are at risk of extinction as a result of climate change.

[4]

[Total: 11]

Essay

Answer **one** question only in this section.

Write your answers on the lined paper provided at the end of this question paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 4 (a) Microorganisms such as *Escherichia coli* (*E. coli*) colonise the intestine and obtain nutrients from their surroundings. [15]
Describe how *E. coli* responds to the presence of lactose and absence of glucose in the intestine and explain how a mutation in the regulatory sequences of the *lac* operon may affect how *E. coli* respond to changes in lactose supply.
- (b) The green fluorescent protein (GFP) gene was isolated from the jellyfish *Aequorea Victoria* and inserted into DNA of a species of mice, *Mus musculus*. The gene was expressed throughout the genetically modified mice resulting in the mice exhibiting bright green fluorescence when exposed to light in the blue to ultraviolet range. [10]
With the use of the various species concepts, discuss whether the genetically modified mice can be considered as the same or different species as the unmodified mice.
- 5 (a) 'Eukaryotic life is impossible without membranes.' [15]
Discuss how regulation of membrane fluidity is achieved in living organisms and justify why the statement above may be true using specific examples.
- (b) With reference to named examples, describe the role of water in reactions involving biomolecules. [10]

[Total: 25]

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